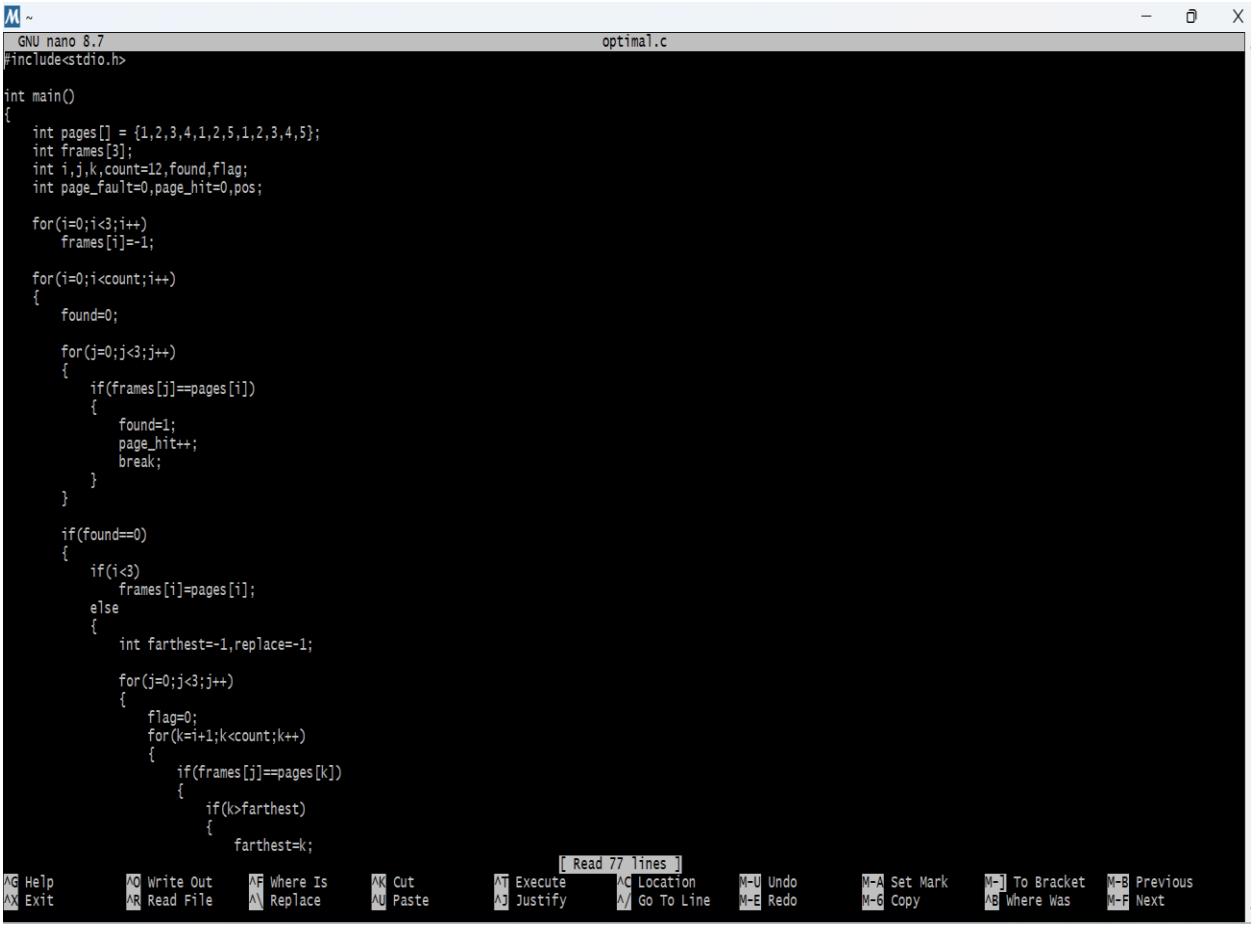
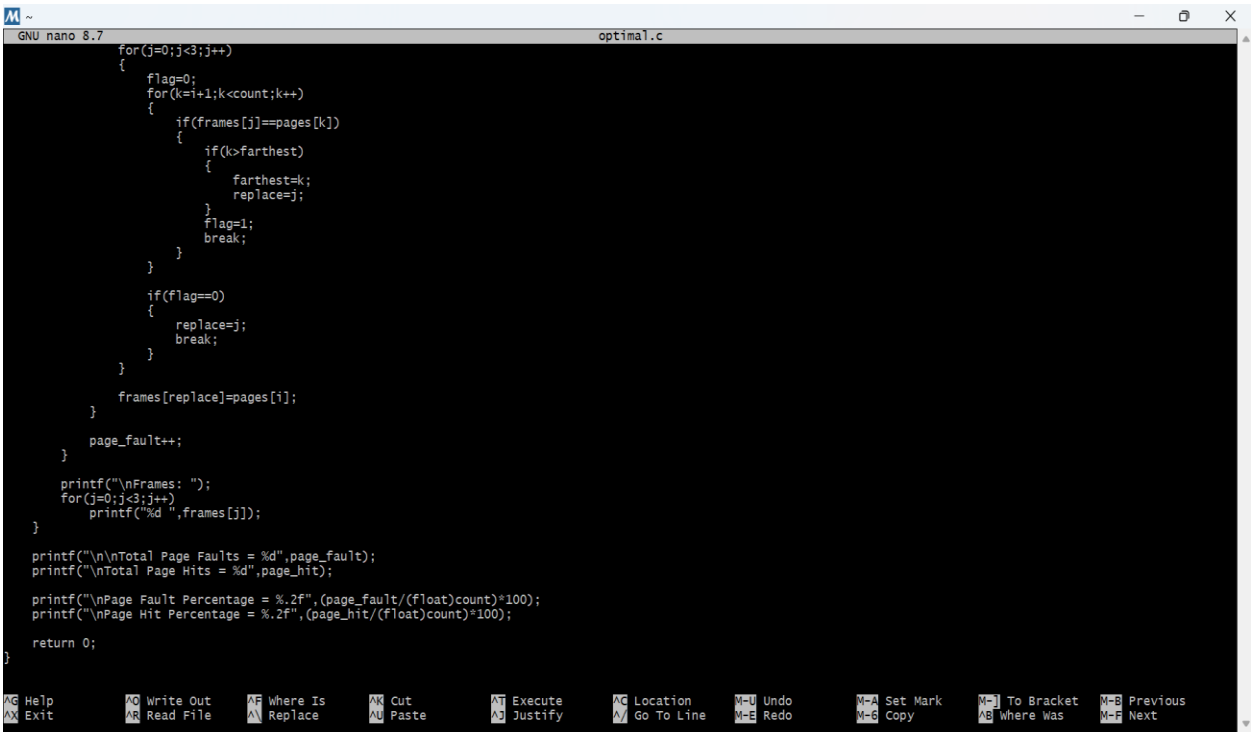
9 [page Replacement Policy ]

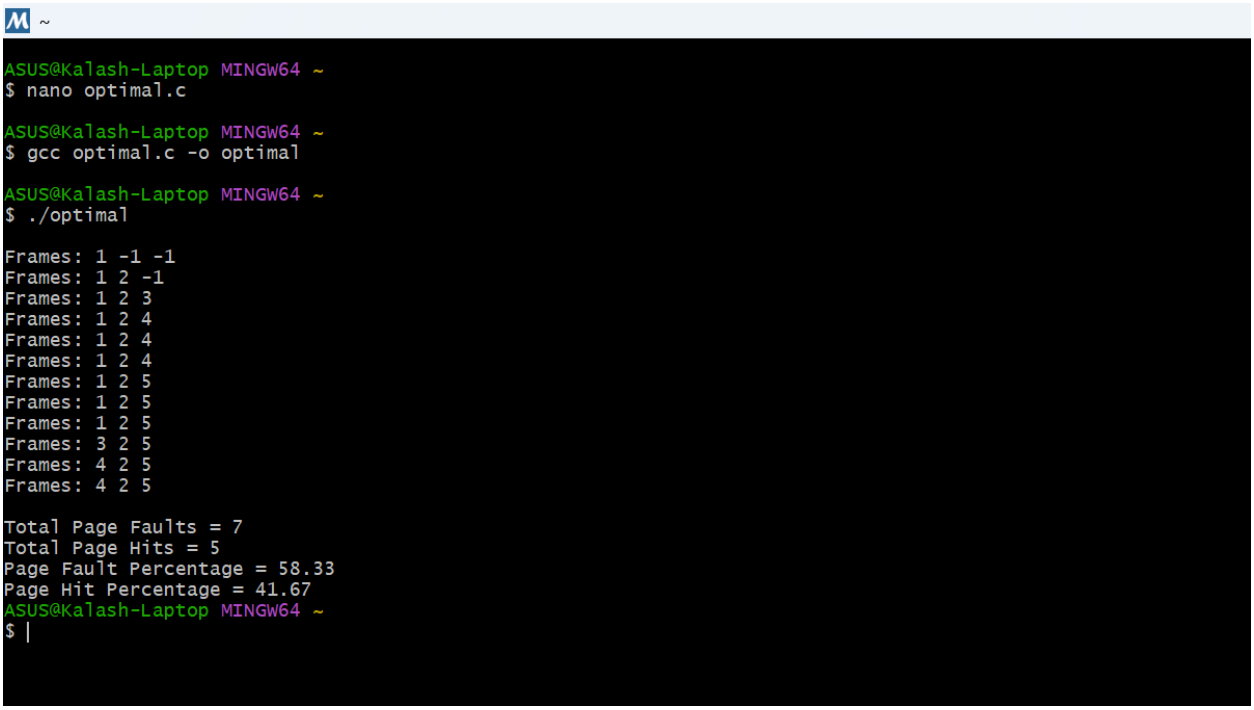
A system has RAM which can store four pages simultaneously to satisfy page requests. Write a Program to calculate percentage page hit and page fault if the system uses Optimal page replacement technique for the page references entered by the user i.e. 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4,5 (No of Frames 3).

**CODE:**




~
GNU nano 8.7 optimal.c
for(j=0;j<3;j++)
{
flag=0;
for(k=i+1;k<count;k++)
{
if(frames[j]==pages[k])
{
if(k>farthest)
{
farthest=k;
replace=j;
}
flag=1;
break;
}
}
if(flag==0)
{
replace=j;
break;
}
}
frames[replace]=pages[i];
}
page_fault++;
}
printf("\nFrames: ");
for(j=0;j<3;j++)
printf("%d ",frames[j]);
}
printf("\n\nTotal Page Faults = %d",page_fault);
printf("\nTotal Page Hits = %d",page_hit);
printf("\nPage Fault Percentage = %.2f",(page_fault/(float)count)*100);
printf("\nPage Hit Percentage = %.2f",(page_hit/(float)count)*100);
return 0;
}
^G Help ^O Write Out ^F Where Is ^K Cut ^T Execute ^C Location M-U Undo M-A Set Mark M-] To Bracket M-B Previous
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line M-E Redo M-6 Copy ^B Where Was M-F Next

## OUTPUT:


~
ASUS@Kalash-Laptop MINGW64 ~
$ nano optimal.c
ASUS@Kalash-Laptop MINGW64 ~
$ gcc optimal.c -o optimal
ASUS@Kalash-Laptop MINGW64 ~
$ ./optimal
Frames: 1 -1 -1
Frames: 1 2 -1
Frames: 1 2 3
Frames: 1 2 4
Frames: 1 2 4
Frames: 1 2 4
Frames: 1 2 5
Frames: 1 2 5
Frames: 1 2 5
Frames: 3 2 5
Frames: 4 2 5
Frames: 4 2 5
Total Page Faults = 7
Total Page Hits = 5
Page Fault Percentage = 58.33
Page Hit Percentage = 41.67
ASUS@Kalash-Laptop MINGW64 ~
$